

$$\equiv x_i + 20 + \left(5\left(\frac{1^2}{x}\right) + 0.00\right) \frac{1.2}{h \bar{c}_j}$$

$$f(x + (5x - x)(0) + 10) = x_i$$

$$2\left(\frac{v}{x}\right)^2 + (x) + \left(\frac{2x}{x}\right) = I_{12}(x) = \frac{2^2}{h} = 5 \frac{2}{x}$$

$$\frac{x(1-x)^{\frac{2}{3}}}{h x^3}$$

$$f(\bar{x}_2)$$

$$x_k + x\left(\frac{1}{h} + x + \left(\frac{kx}{m^2}\right) \mp 1\right)$$

$$= \frac{x}{x_j} + 0 - (x^2 + 0) = \frac{x}{l_x a m^{x_k}}$$

$$f(0, +5, +x) = (1, 3)$$

# Engineering Mathematics III: Second Order NonHomogenous ODEs

Week 4 Lecture 4

# Housekeeping

## 1 Pre-class recordings

Available on Canvas - access both sections

## 2 Assignment 1

Quizzes 1 to 6

Additional component



# The Forced Harmonic Oscillator

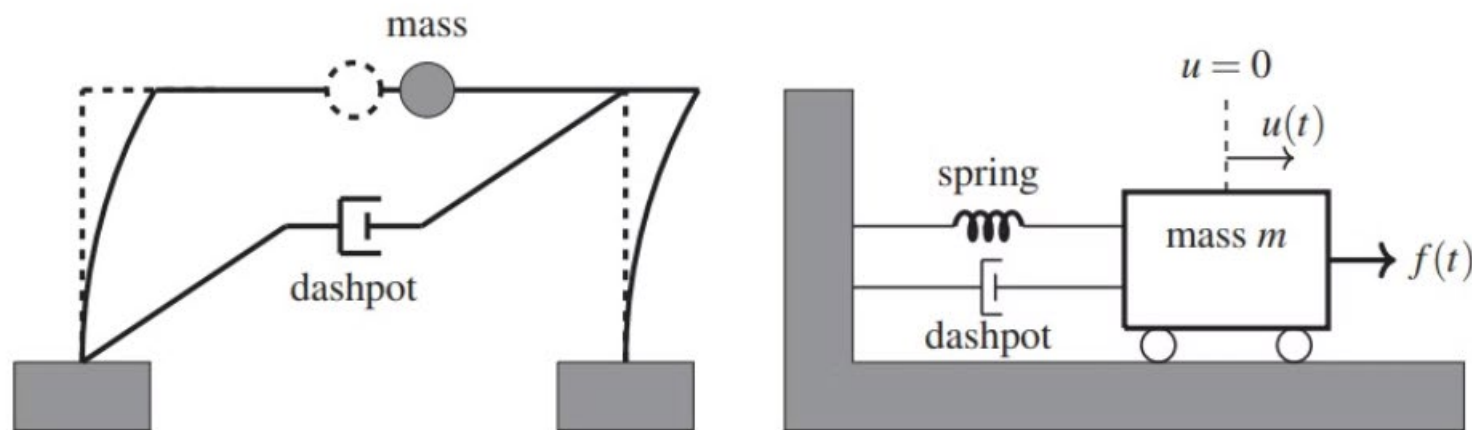
$$y'' + p(x)y' + q(x)y = r(x) \text{ where } r(x) \neq 0$$

The unforced harmonic oscillator  $y'' + p(x)y' + q(x)y = 0$

stems from the assumption that the only forces acting on the mass  $m$  are the spring force  $F_{spring}$  ( $F_{spring} = -q(x)y$ ) and frictional forces of the form  $F_{damping}$  ( $F_{damping} = -p(x)y'$ ).

However, it is often the case that additional forces act on the mass.

For example, in our simplified earthquake model from last week, the shaking of the ground is the very thing that sets the mass in motion.



# Nonhomogeneous Forced Harmonic Oscillator ODE

$$y'' + p(x)y' + q(x)y = r(x) \text{ where } r(x) \neq 0$$

In this lecture we will look at methods of solving second order nonhomogeneous linear ODEs.

Finding a solution to this type of differential equation is very much related to solving the corresponding homogeneous equation (ie with  $r(x) = 0$ ).

The **solution** will in fact be the **solution to the corresponding homogeneous equation, plus a particular solution to the original, nonhomogeneous equation.**

As in the homogeneous case  $y'' + p(x)y' + q(x)y = 0$  two initial conditions are needed to specify a unique solution.

The two-fold focus in this lecture is the general structure of solutions to  $y'' + p(x)y' + q(x)y = r(x)$  where  $r(x) \neq 0$  and how to find solutions with specified initial conditions. But first, let's look at an example.

# Example: Spring-Mass-Damper System

Consider our simplified earthquake model that we will model as a spring-mass-damper system with  $m = 5000$  kg,  $k = 5 \times 10^5$  newtons per meter, and  $c = 10^4$  newton-seconds per meter.

With no additional forces on the mass the relevant ODE is  $5000u'' + 10^4u' + (5 \times 10^5)u = 0$ .

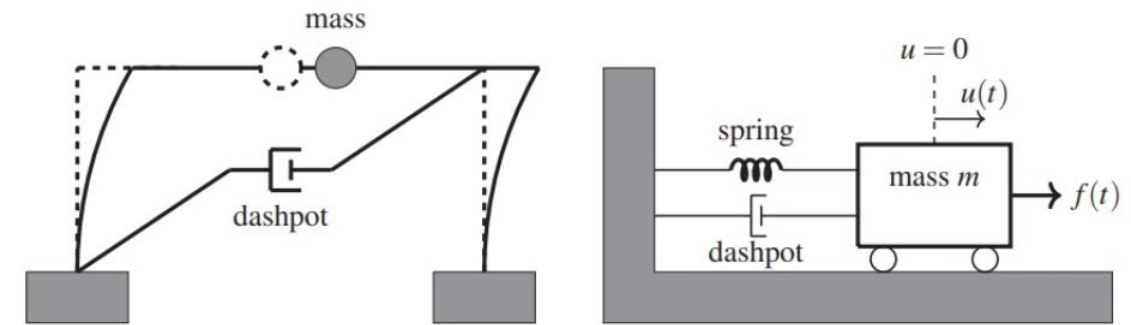
The roots to the characteristic equation are *approximately*  $-1 \pm 9.95i$

This gives us a general solution is

$$u(t) = c_1 e^{-t} \cos(9.95t) + c_2 e^{-t} \sin(9.95t)$$

With initial conditions  $u(0) = 0.01$  and  $u'(0) = 0$  the solution is

$$u(t) = 0.01 e^{-t} \cos(9.95t) + 0.001 e^{-t} \sin(9.95t)$$



## System Parameters

- Mass:  $m = 5000$  kg
- Spring constant:  $k = 5 \times 10^5$  newtons per meter
- Damping coefficient:  $c = 10^4$  newton-seconds per meter

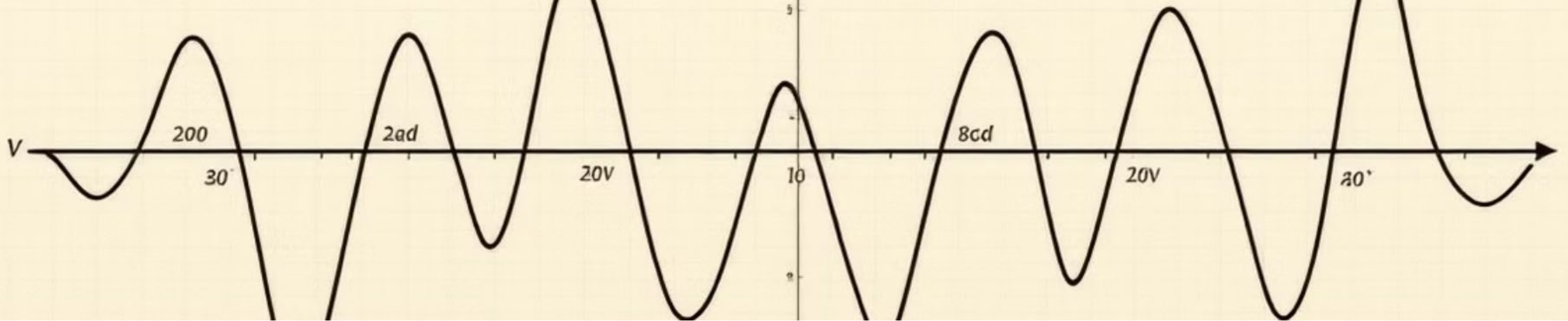
Based on Bryan (2022) 4.3

## Resulting ODE

$$5000u'' + 10^4u' + (5 \times 10^5)u = 0$$

## General Solution Homogeneous equation

$$u(t) = c_1 e^{-t} \cos(9.95t) + c_2 e^{-t} \sin(9.95t)$$



## Case 3: Complex Conjugate Roots

$$y'' + ay' + by = 0$$

$$\lambda^2 + a\lambda + b = 0$$

### 1 Condition

When  $a^2 - 4b < 0$ , the characteristic equation has two complex conjugate roots  $\lambda = -a/2 \pm i\omega$ , where  $\omega^2 = b - (a^2/4)$ .

### 2 Solutions

The linearly independent solutions are

$$y_1 = e^{\frac{-ax}{2}} \cos(\omega x)$$

and

$$y_2 = e^{\frac{-ax}{2}} \sin(\omega x).$$

### 3 General Solution

The general solution is

$$y = e^{\frac{-ax}{2}} (A \cos(\omega x) + B \sin(\omega x)),$$

where A and B are arbitrary constants.

# Summary of Solution Methods

| Case | Roots                                                                          | General Solution                                        |
|------|--------------------------------------------------------------------------------|---------------------------------------------------------|
| 1    | Distinct real roots $\lambda_1, \lambda_2$                                     | $y = c_1 e^{\lambda_1 x} + c_2 e^{\lambda_2 x}$         |
| 2    | One real root $\lambda$                                                        | $y = (c_1 + c_2 x) e^{\lambda x}$                       |
| 3    | Complex conjugate roots $-a/2 \pm i\omega$<br>where $\omega^2 = b - (a^2/4)$ . | $y = e^{(-ax/2)} (A \cos(\omega x) + B \sin(\omega x))$ |

The solution method for second order homogeneous ODEs with constant coefficients can be summarized in these three cases.

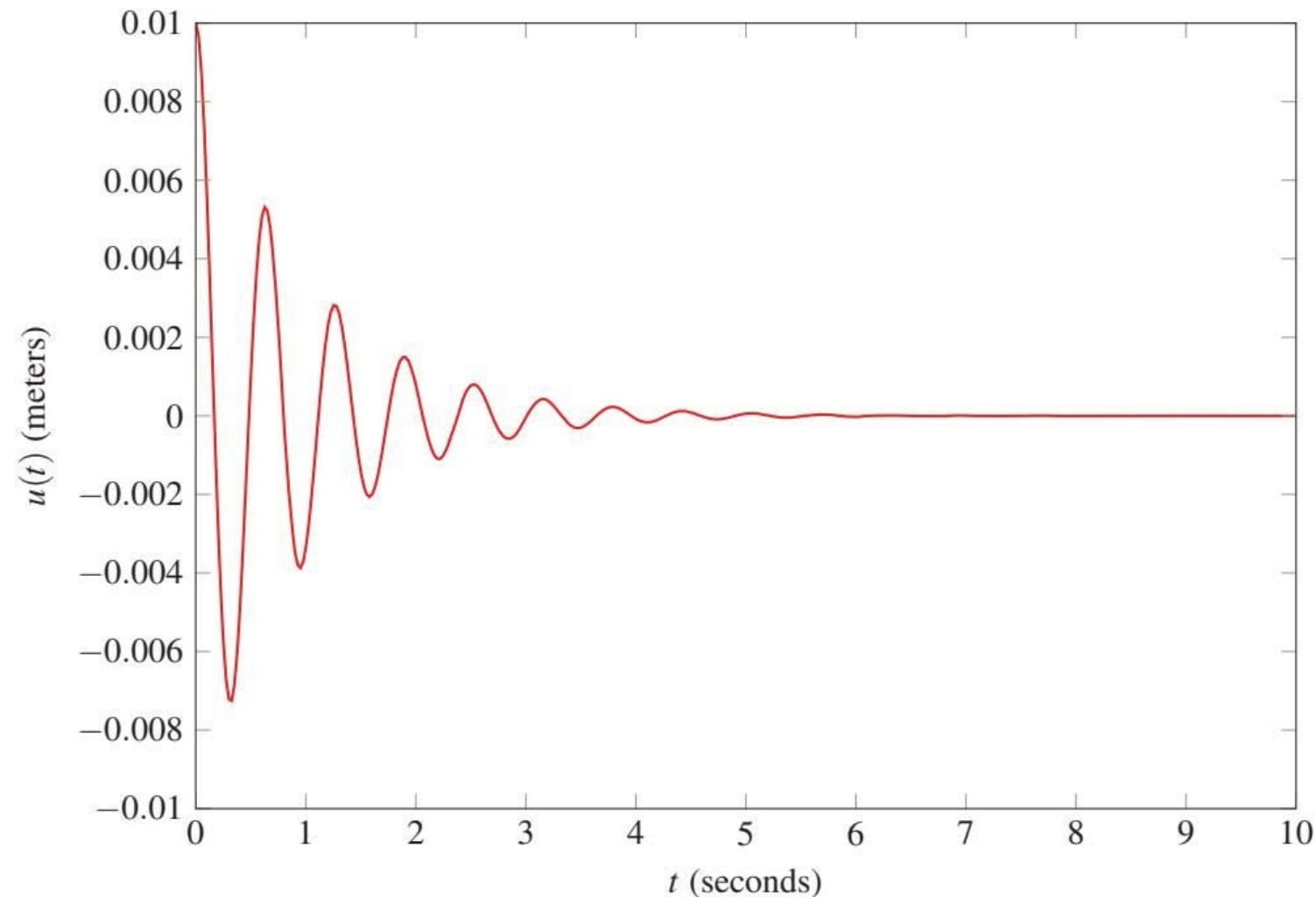
The type of roots in the characteristic equation determines which formula to use for the general solution.

# Example: Spring-Mass-Damper System ...

$u(t) = 0.01e^{-t} \cos(9.95t) + 0.001e^{-t} \sin(9.95t)$  is plotted in the Figure below.

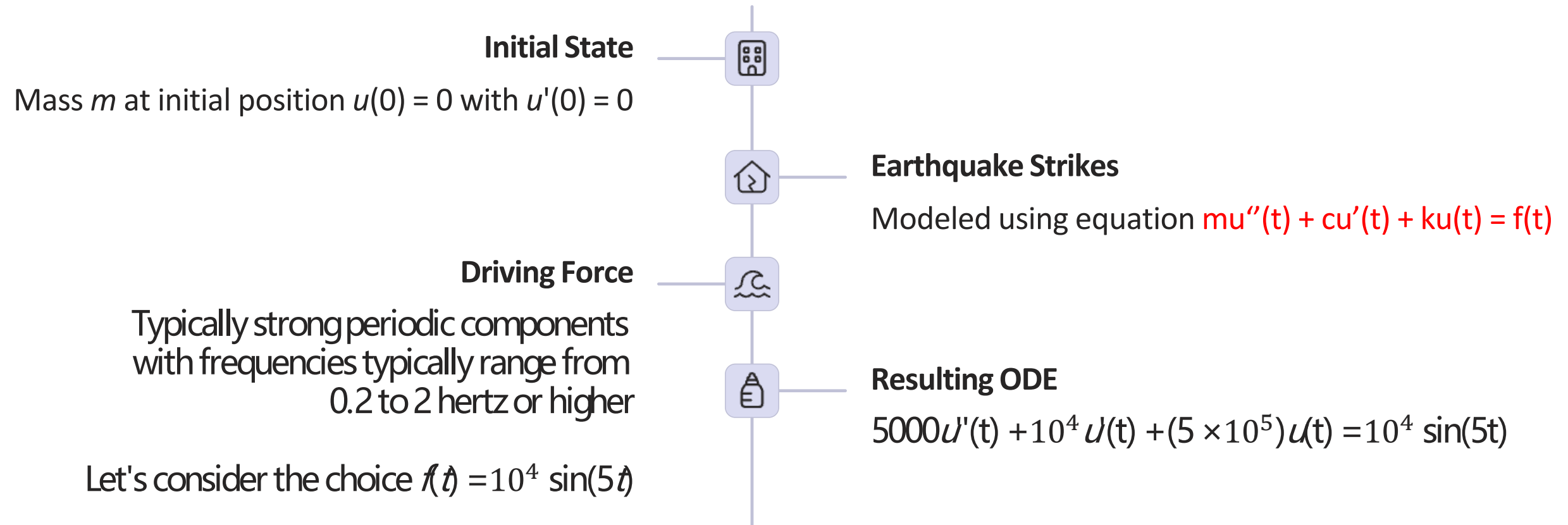
As expected, since there are no external forces acting on the system after time  $t > 0$ , the oscillations damp out after a few cycles (though there may have been external forces for  $t \leq 0$ , in order to initiate the initial nonzero displacement).

The sinusoidal portion of the damped oscillations has a frequency of  $9.95/2\pi \approx 1.58$  hertz, or a period of about 0.63 seconds.



How would the building respond in an earthquake?

# Modeling Earthquake Effects



Here we will model an earthquake as a driving force on the system (but see the corresponding Project "Earthquake Modeling" in "Earthquake Modeling" in Section 4.6 Bryan (2022) for a slightly more realistic model.)

The response to the driven system may be quite different from the undriven system and depend very much on the nature of the the nature of the driving force.

# Solution Analysis

## Transient Component

$$0.0128e^{-t} \sin(9.95t) + 0.0035e^{-t} \cos(9.95t)$$

- Contains  $e^{-t}$  terms
- Quickly decays to zero
- Temporary response

## Periodic Component

$$0.02628 \sin(5t) - 0.0035 \cos(5t)$$

- Does not decay with time
- Persists as long as force is active
- Dominates the solution as  $t$  increases
- Same frequency as driving force

Here  $f(t)$  is periodic with frequency  $5/(2\pi) \approx 0.8$  hertz and amplitude 104 newtons, which is about 2040 pounds. In this case the solution to the ODE with  $u(0) = u'(0) = 0$  is, to three significant figures,

$$u(t) = 0.0128e^{-t} \sin(9.95t) + 0.0035e^{-t} \cos(9.95t) + 0.02628 \sin(5t) - 0.0035 \cos(5t)$$

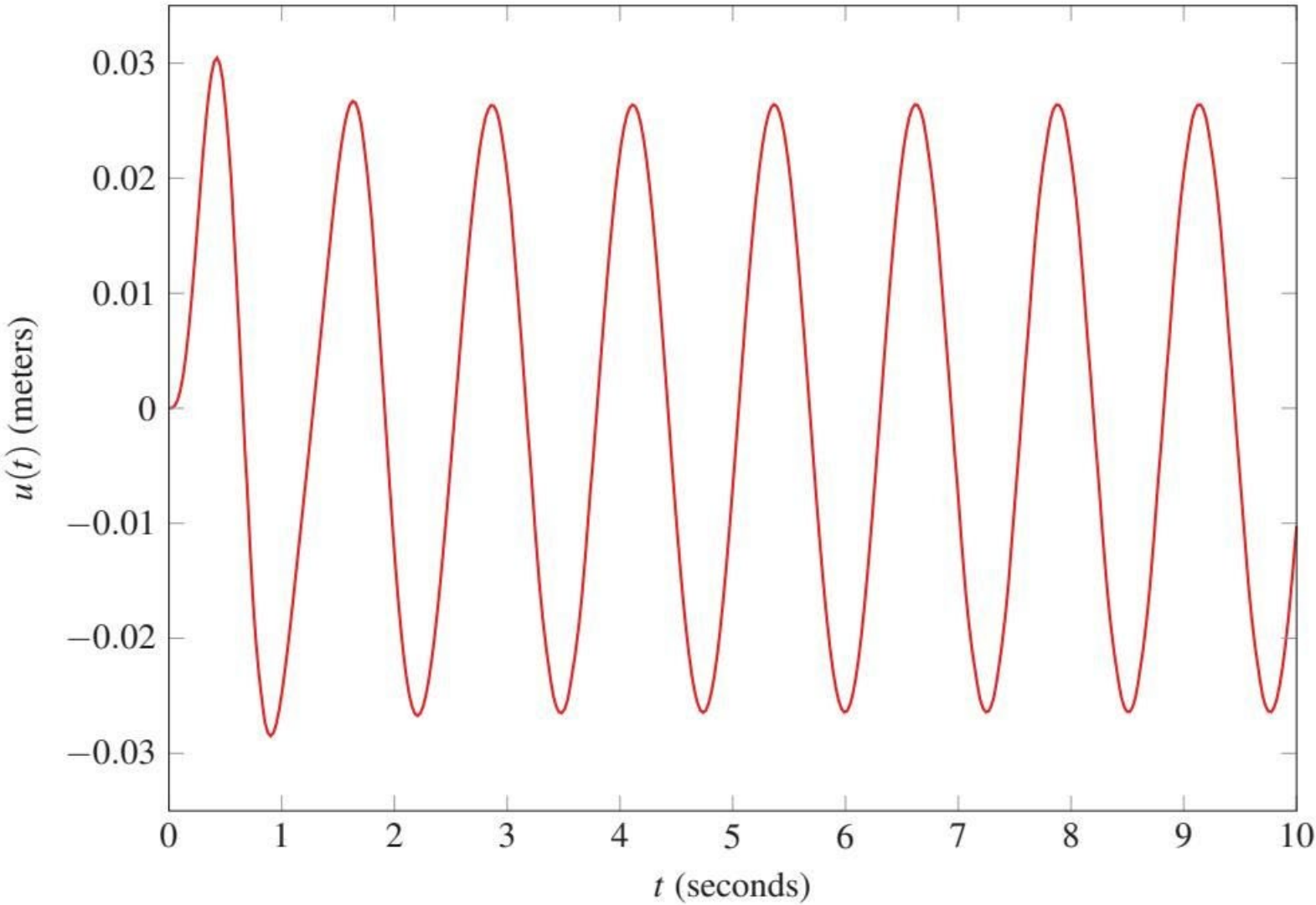
The solution  $u(t)$  has a **transient portion** stemming from the terms that contains  $e^{-t}$ .

This part of the solution quickly decays to zero.

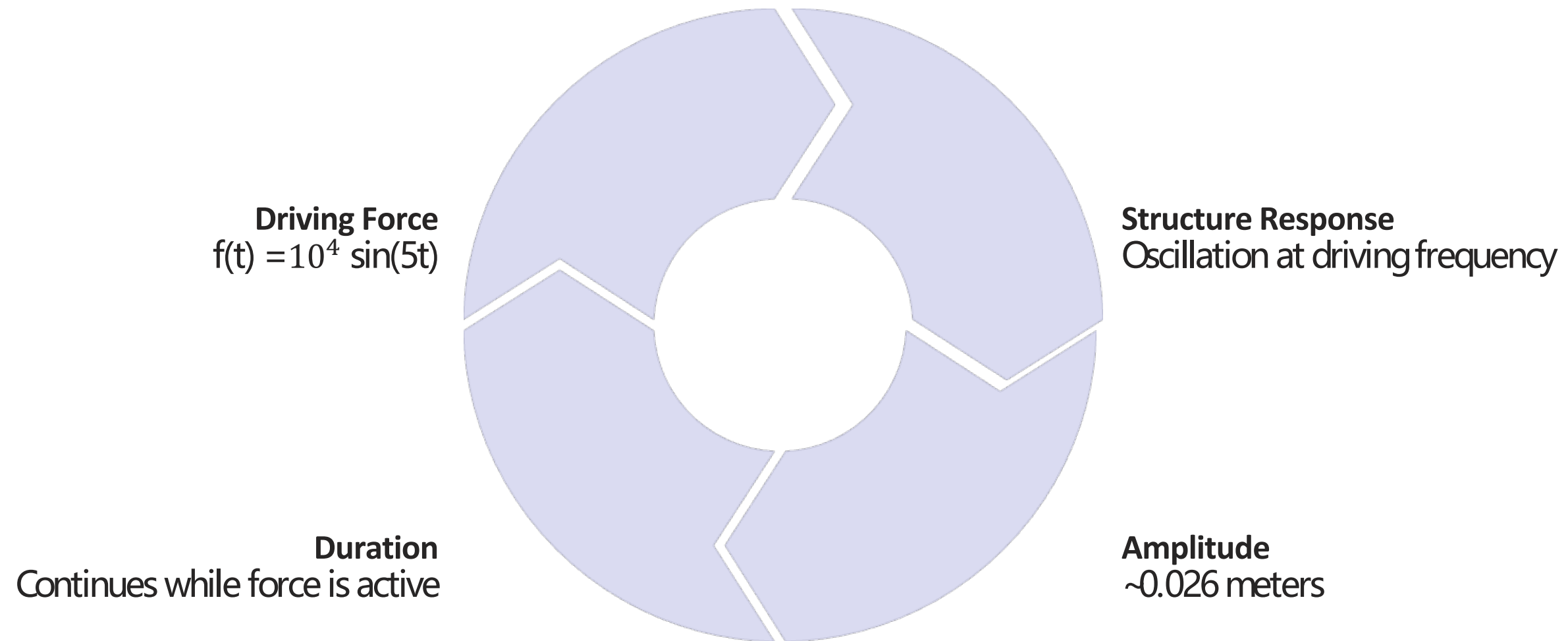
However, the two terms on the right in  $u(t)$  involving  $\sin(5t)$  and  $\cos(5t)$  are the **periodic portion** of the solution and do not decay in time. The periodic portion persists, and quickly dominates the solution as  $t$  increases. The long-term response of the building is to shake sinusoidally at the same frequency as the driving force  $f(t)$ , with an amplitude of about 0.026 meters.

This response will continue as long as  $f(t)$  remains active.

# Response Visualization



$$u(t) = 0.0128e^{-t} \sin(9.95t) + 0.0035e^{-t} \cos(9.95t) + 0.02628 \sin(5t) - 0.0035 \cos(5t)$$



**Response  $u(t)$  of the forced oscillator**  $5000u'(t) + 10^4 u(t) + (5 \times 10^5)u(t) = 10^4 \sin(5t)$

# General Solution Structure

## Complete Solution

Given a second order linear differential equation

$$y'' + p(x)y' + q(x)y = r(x),$$

the general solution will be of the form

$$y = y_p + y_h$$

## Homogeneous Solution

$y_h$  is the general solution to the homogeneous equation

$$y'' + p(x)y' + q(x)y = 0$$

## Particular Solution

$y_p$  is any particular solution to the original equation

$$y'' + p(x)y' + q(x)y = r(x)$$

Now we have two distinct jobs to do to solve the nonhomogeneous equation.

We already know how to solve second order homogeneous linear differential equations with constant coefficients.

Now we will cover a method to find a particular solution to a **second order nonhomogeneous linear differential equation with constant coefficients**.

# Method of Undetermined Coefficients



## The Approach

We already know how to solve second order homogeneous differential equations with constant coefficients, we will see now how to find a particular solution using a method known as the **method of undetermined coefficients**.



## The Method

The method of undetermined coefficients works by 'guessing' a solution (with some unknown coefficients) based on  $r(x)$ , and then putting this back into the equation to work out what the coefficients should be.



## Reference Table

We have a table of 'guesses' for common forms of  $r(x)$  provides a structured approach to finding particular solutions for different types of functions.

# Common Forms for Particular Solutions - Table of ‘guesses’

| Term in r(x)                  | Choice for $y_p$                                      |
|-------------------------------|-------------------------------------------------------|
| $ke^{yx}$                     | $Ke^{yx}$                                             |
| $kx^n$                        | $Kx^n + K_{n-1}x^{n-1} + \dots + K_1x + K_0$          |
| $k\cos(\omega x)$             | $K_1\cos(\omega x) + K_2\sin(\omega x)$               |
| $k\sin(\omega x)$             | $K_1\cos(\omega x) + K_2\sin(\omega x)$               |
| $ke^{\alpha x}\cos(\omega x)$ | $e^{\alpha x}(K_1\cos(\omega x) + K_2\sin(\omega x))$ |
| $ke^{\alpha x}\sin(\omega x)$ | $e^{\alpha x}(K_1\cos(\omega x) + K_2\sin(\omega x))$ |

This table provides a systematic approach to selecting the appropriate form for the particular solution based on the form of r(x) in the original differential equation.

Once the form is selected, the undetermined coefficients can be found by substituting the proposed solution into the original equation.

# Method of Undetermined Coefficients

| Term in $r(x)$                | Choice for $y_p$                                      |
|-------------------------------|-------------------------------------------------------|
| $ke^{yx}$                     | $Ke^{yx}$                                             |
| $kx^n$                        | $Kx^n + K_{n-1}x^{n-1} + \dots + K_1x + K_0$          |
| $k\cos(\omega x)$             | $K_1\cos(\omega x) + K_2\sin(\omega x)$               |
| $k\sin(\omega x)$             | $K_1\cos(\omega x) + K_2\sin(\omega x)$               |
| $ke^{\alpha x}\cos(\omega x)$ | $e^{\alpha x}(K_1\cos(\omega x) + K_2\sin(\omega x))$ |
| $ke^{\alpha x}\sin(\omega x)$ | $e^{\alpha x}(K_1\cos(\omega x) + K_2\sin(\omega x))$ |

Using the table, we can choose an appropriate  $y_p$  using the following rules:

- **Basic Rule:** If  $r(x)$  is one of the functions in the first column of the table on the previous slide, choose the corresponding  $y_p$ , then determine the coefficients by inserting  $y_p$  and its derivatives into the differential equation, and equating the sides.
- **Sum Rule:** If  $r(x)$  is a sum of functions in the first column of the table on the previous slide, choose  $y_p$  to be the sum of the corresponding functions in the table, and then determine the coefficients by inserting  $y_p$  and its derivatives into the differential equation, and equating the sides.
- **Modification Rule:** If a term in the choice for  $y_p$  is a solution of the corresponding homogeneous ODE, then multiply this term by  $x$  (or  $x^2$  if the solution is a double root).

# Find the general solution to $y'' + y' - 2y = \sin(x)$

## Identify the Problem

Find the general solution to the differential equation  $y'' + y' - 2y = \sin(x)$ .

The solution will be the sum of  $y_p$  and  $y_h$ .

## Find the Homogeneous Solution

We know the solution of the corresponding homogeneous equation is  $y_h = c_1 e^x + c_2 e^{-2x}$

## Choose Form for Particular Solution

From the table, we see that the suggested choice for  $y_p = K_1 \cos(x) + K_2 \sin(x)$

## Find Derivatives

Then  $y_p' = K_2 \cos(x) - K_1 \sin(x)$  and  $y_p'' = -K_1 \cos(x) - K_2 \sin(x)$

## Termin r(x)

$$k \sin(\omega x)$$

## Choice for $y_p$

$$K_1 \cos(\omega x) + K_2 \sin(\omega x)$$

# Find the general solution to $y'' + y' - 2y = \sin(x)$ ...

## Substitute into Original Equation

$$\begin{aligned} y_p'' + y_p' - 2y_p &= -K_1 \cos(x) - K_2 \sin(x) + K_2 \cos(x) - K_1 \sin(x) - 2K_1 \cos(x) - 2K_2 \sin(x) \\ &= (K_2 - 3K_1) \cos(x) + (-K_1 - 3K_2) \sin(x) \end{aligned}$$

## Solve for Coefficients

Since this must equate to  $0 \cos(x) + 1 \sin(x)$ , as

$$(K_2 - 3K_1) \cos(x) + (-K_1 - 3K_2) \sin(x) = \sin(x)$$

$$K_2 - 3K_1 = 0$$

we conclude that

and

$$-K_1 - 3K_2 = 1$$

Solving this we get

$$K_1 = -\frac{1}{10} \quad \text{and} \quad K_2 = -\frac{3}{10}$$

## Write the Complete Solution

Then our particular solution is  $y_p = -\frac{1}{10} \cos(x) - \frac{3}{10} \sin(x)$ , so our general solution is then

$$y = y_h + y_p = c_1 e^x + c_2 e^{-2x} - \frac{1}{10} \cos(x) - \frac{3}{10} \sin(x)$$

# Find the general solution to $y'' - y' - 2y = 2e^{-x}$

$f(x)$

## Identify the Problem

Find the general solution to the differential equation  $y'' - y' - 2y = 2e^{-x}$



## Find the Homogeneous Solution

Using what we know about solving homogeneous equations, we can see that  $y_h = c_1 e^x + c_2 e^{2x}$



## Choose Form for Particular Solution

From the table, we see that the suggested choice for  $y_p$  is  $y_p = Ce^{-x}$ . This is the same as one of our solutions, so instead we use  $y_p = Cxe^{-x}$ .



## Find Derivatives

Then  $y'_p = C(1 - x)e^{-x}$  and  $y''_p = C(x - 2)e^{-x}$ .



## Substitute into Original Equation

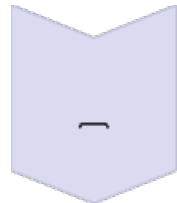
$$\begin{aligned} y''_p - y'_p - 2y_p &= C(x - 2)e^{-x} - C(1 - x)e^{-x} - 2Cxe^{-x} \\ &= -3Ce^{-x} \end{aligned}$$

# Find the general solution to $y'' - y' - 2y = 2e^{-x}$



## Solve for Coefficient

Equating this to  $r(x)$  we get  $-3C = 2$ , so  $C = -2/3$ .



## Write the Complete Solution

Then our general solution is

$$y = c_1 e^{-x} + c_2 e^{2x} - \frac{2}{3} x e^{-x}$$

# Practice Exercises

## Exercise 1

Find the solution to the following differential equation:

$$y'' + y = e^{(2x)}$$

## Exercise 2

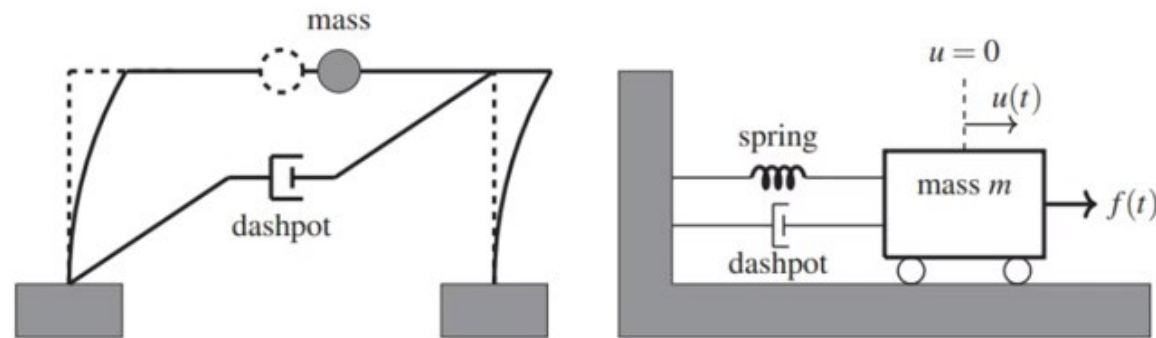
Find the solution to the following differential equation:

$$y'' - y = x^2$$

**Approach** For each exercise:

1. Find the homogeneous solution  $y_h$
2. Determine the form of the particular solution  $y_p$
3. Find the coefficients for  $y_p$
4. Combine to get the general solution  $y = y_h + y_p$

# Review: Mass-Spring-Damper System System with External Force



:



## Spring Component

Provides restoring force proportional to displacement



## Mass Component

Responds to forces according to Newton's laws



## Damper Component

Provides resistance proportional to velocity



## External Force

Additional force acting on the system

# System Response Analysis

## Input Force

External periodic force applied to the system  
system

## Resonance

Maximum amplitude when forcing frequency approaches  
approaches natural frequency



## System Dynamics

Mass, spring, and damper properties  
determine response

## Response Amplitude

Depends on relationship between forcing  
forcing and natural frequencies

# Periodic External Force

## 1 Periodic Force Equation

The most interesting case is when  $r(t)$  is periodic, and we have the equation

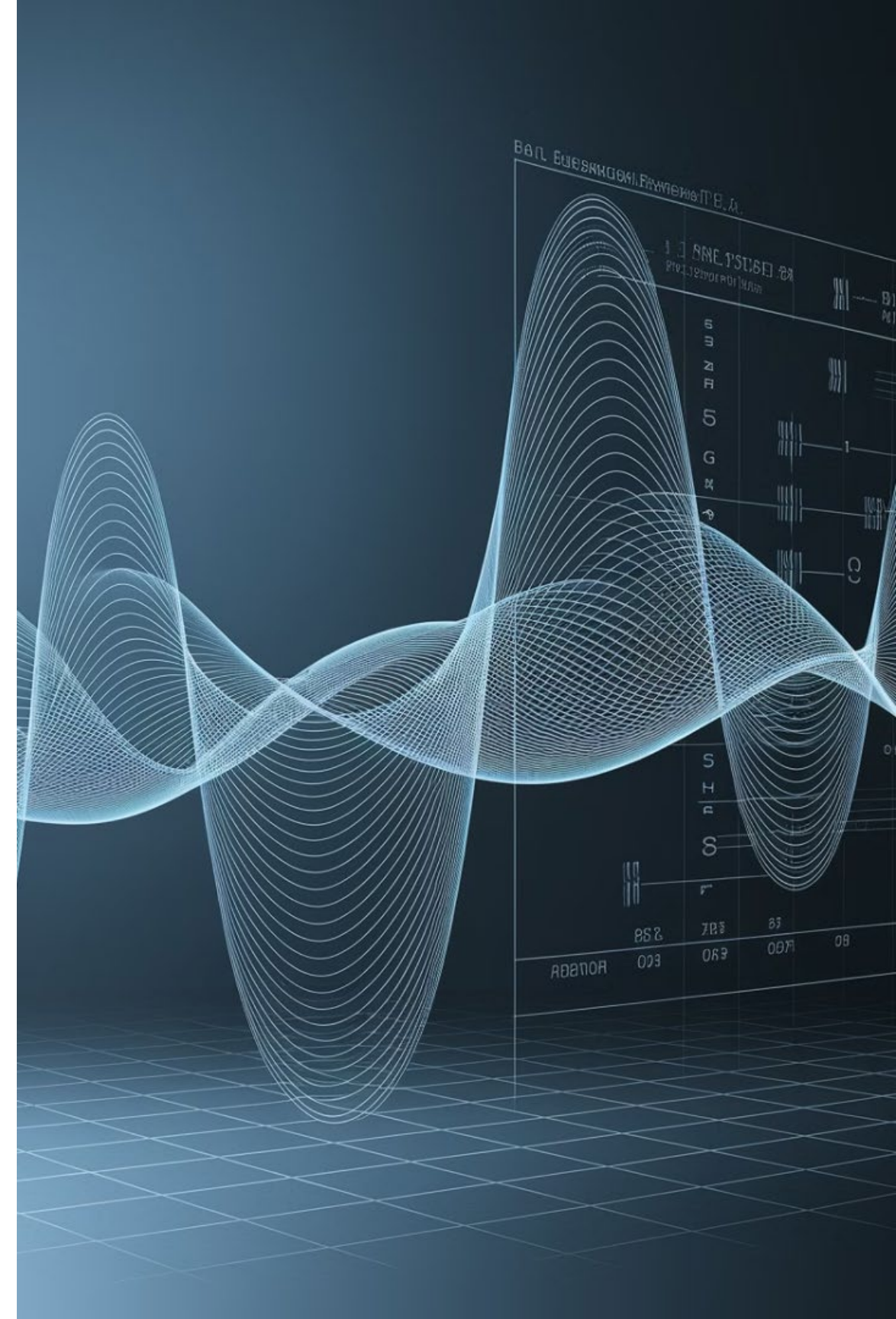
$$my'' + cy' + ky = F_0 \cos(\omega t).$$

## 3 Homogeneous Solution

The solution  $y_h$  will depend on whether the system is not damped, underdamped, overdamped, or neutrally damped.

## 2 Solution Approach

We can find the solution to this as the sum of the corresponding homogeneous equation solution  $y_h$  and a particular solution  $y_p$ .



A man in a white shirt and glasses is holding a transparent tablet. The tablet displays various mathematical formulas and tables. At the top, it shows  $y'' + cy' + ky = F_0 \cos(\omega t)$ . Below this, there are several tables with columns and rows containing numbers and symbols. The background is a blurred image of a modern building with geometric patterns.

# Finding the Particular Solution

$$my'' + cy' + ky = F_0 \cos(\omega t).$$

## Identify Solution Form

From the table of 'guesses', we know that our particular solution will be of the form

$$y_p = a \cos(\omega t) + b \sin(\omega t).$$

## Take Derivatives

From here, we can take derivatives of  $y_p$ , and put it into our differential equation.

## Solve for Constants

Solve for  $a$  and  $b$  by substituting into the original differential equation.

# Solution Coefficients

Solving for a and b we get:

## Coefficient a

$$a = F_0 \frac{m(\omega_0^2 - \omega^2)}{m^2(\omega_0^2 - \omega^2)^2 + \omega^2 c^2}$$

## Coefficient b

$$b = F_0 \frac{\omega c}{m^2(\omega_0^2 - \omega^2)^2 + \omega^2 c^2}$$

## Natural Frequency

$\omega_0 = \sqrt{k/m}$  is called the 'natural frequency'.

$$\omega_0 = \sqrt{\frac{k}{m}}$$



$$my'' + cy' + ky = F_0 \cos(\omega t).$$

Assume  $y_p = a \cos(\omega t) + b \sin(\omega t)$

$$\left\{ \begin{array}{l} y_p' = -a\omega \sin(\omega t) + b\omega \cos(\omega t) \\ y_p'' = -a\omega^2 \cos(\omega t) - b\omega^2 \sin(\omega t) \end{array} \right.$$

$$-a\omega^2 \cos(\omega t) - b\omega^2 \sin(\omega t) - a c \omega \sin(\omega t) + b c \omega \cos(\omega t) + k a \cos(\omega t) + k b \sin(\omega t) = F_0 \cos(\omega t)$$

$$\Rightarrow (-a\omega^2 + b c \omega + a k) \cos(\omega t)$$

$$+ (-b\omega^2 - a c \omega + b k) \sin(\omega t) = F_0 \cos(\omega t) + 0 \sin(\omega t)$$

$$-a\omega^2 + b c \omega + a k = F_0$$

$$-b\omega^2 - a c \omega + b k = 0$$

$$-am\omega^2 + bc\omega + ak = F_0$$

$$-bm\omega^2 - ac\omega + bk = 0$$

$$\Rightarrow a = \frac{bk - bm\omega^2}{c\omega}$$

$$bc\omega - b \frac{(k - m\omega^2)^2}{c\omega} = F_0$$

$$b = \frac{F_0 c\omega}{c^2\omega^2 - (k - m\omega^2)^2}$$

$$a = b \frac{(k - m\omega^2)}{c\omega} = \frac{F_0 (k - m\omega^2)}{c^2\omega^2 - (k - m\omega^2)^2}$$

$$a = F_0 \frac{m(\omega_0^2 - \omega^2)}{m^2(\omega_0^2 - \omega^2)^2 + \omega^2 c^2} \quad \text{and} \quad b = F_0 \frac{\omega c}{m^2(\omega_0^2 - \omega^2)^2 + \omega^2 c^2}$$

$$\omega_0 = \sqrt{\frac{k}{m}}$$

$$y_p = a \cos(\omega t) + b \sin(\omega t)$$

# Case 1: Undamped Forced Oscillations

$$my'' + cy' + ky = F_0 \cos(\omega t).$$

$$\omega_0 = \sqrt{\frac{k}{m}}$$

$$y_p = a \cos(\omega t) + b \sin(\omega t)$$

$$a = F_0 \frac{m(\omega_0^2 - \omega^2)}{m^2(\omega_0^2 - \omega^2)^2 + \omega^2 c^2}$$

$$b = F_0 \frac{\omega c}{m^2(\omega_0^2 - \omega^2)^2 + \omega^2 c^2}$$

$$y_h = (A \cos(\omega t) + B \sin(\omega t))$$

1

**Set Damping Coefficient  $c = 0$**

This simplifies our equations for the undamped case

2

**Coefficients**

$$a = \frac{F_0}{m(\omega_0^2 - \omega^2)} \quad \text{and} \quad b = 0$$

3

**Particular Solution**

$$y_p = \frac{F_0}{m(\omega_0^2 - \omega^2)} \cos(\omega t)$$

4

**General Solution**

$$y = A \sin(\omega_0 t) + B \cos(\omega_0 t) + \frac{F_0}{m(\omega_0^2 - \omega^2)} \cos(\omega t)$$

In undamped forced oscillations, we set the damping coefficient  $c$  to zero. This simplifies our equations considerably, with coefficient  $b$  becoming zero. The particular solution takes the form of a cosine function with an amplitude that depends on the difference between the natural and forcing frequencies.

The general solution combines both the homogeneous solution (first two terms) and the particular solution (third term). This equation describes how an undamped system responds to a periodic forcing function.

$$my'' + cy' + ky = F_0 \cos(\omega t).$$

$$\omega_0 = \sqrt{\frac{k}{m}}$$

We will now look at what happens when  $\omega = \omega_0$ . In this case, one of the terms in our homogeneous solution  $y_h$  contains a  $\cos(\omega_0 t)$  term, as does the suggested particular solution. So we need to use the modification rule, and our suggested particular solution becomes:

$$y_p = t(a \cos(\omega_0 t) + b \sin(\omega_0 t))$$

This gives (after finding the constants  $a$  and  $b$ ):

$$y_p = \frac{F_0}{2m\omega_0} t \sin(\omega_0 t)$$

.

This gives displacements looking something like this:

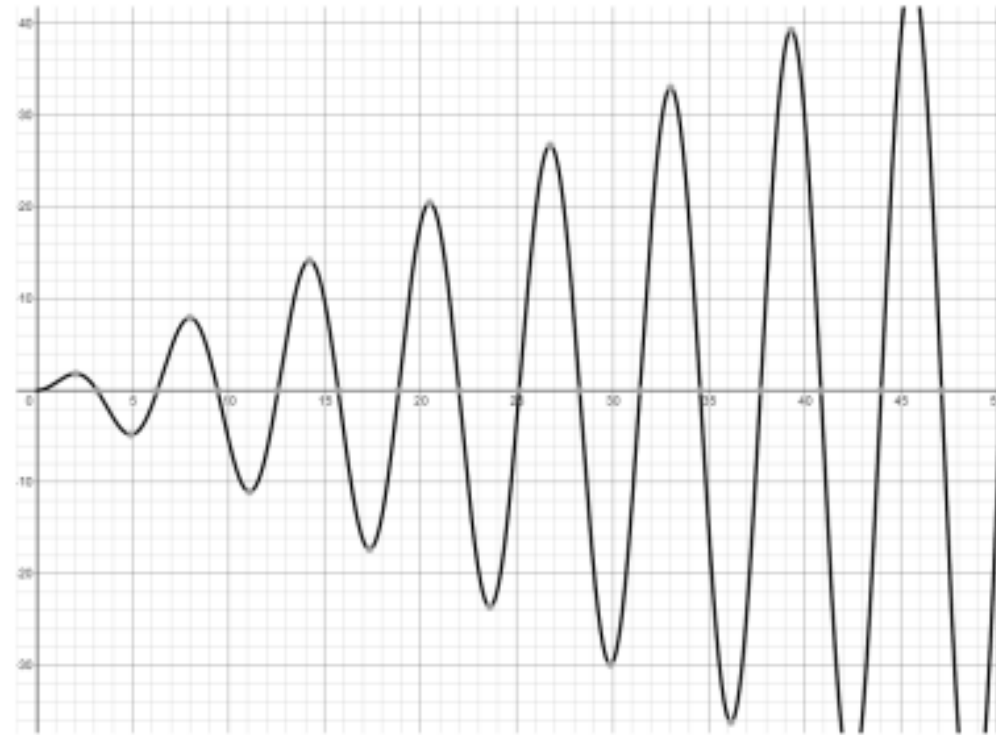


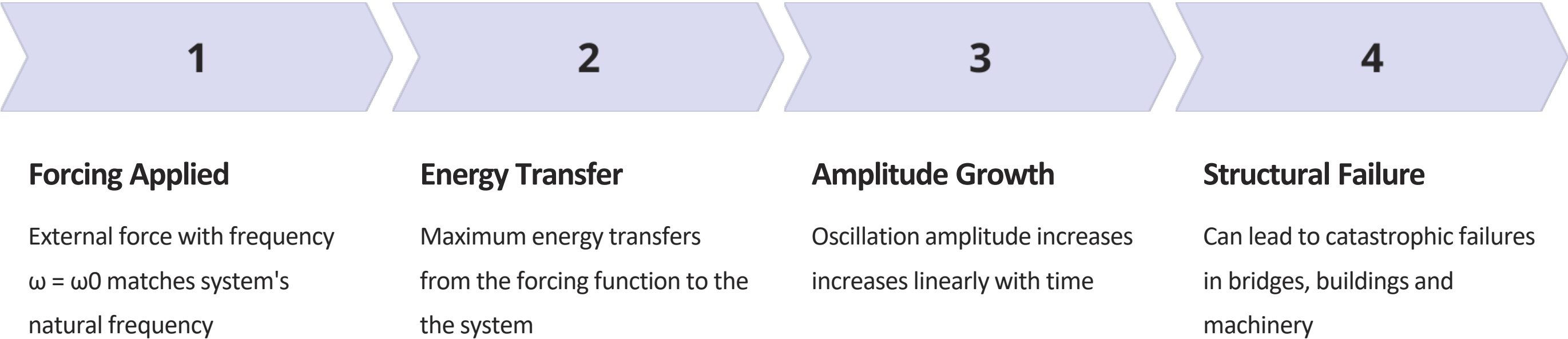
Figure: Resonance

In practical terms, this means the system would likely destroy itself, unless it caused it to be in a state where our model no longer applies.

# Resonance Phenomenon

When the forcing frequency equals the natural frequency ( $\omega = \omega_0$ ), we observe resonance.

The particular solution becomes proportional to  $t \cdot \sin(\omega_0 t)$ , causing amplitude to grow unbounded over time.



This explains why soldiers break step when crossing bridges and why washing machines need balancing.

Another similar phenomenon is known as **beats**. Here the driving frequency is 'close' to the natural frequency.

We showed earlier that the general solution will be

$$y = A \sin(\omega_0 t) + B \cos(\omega_0 t) + \frac{F_0}{m(\omega_0^2 - \omega^2)} \cos(\omega t)$$

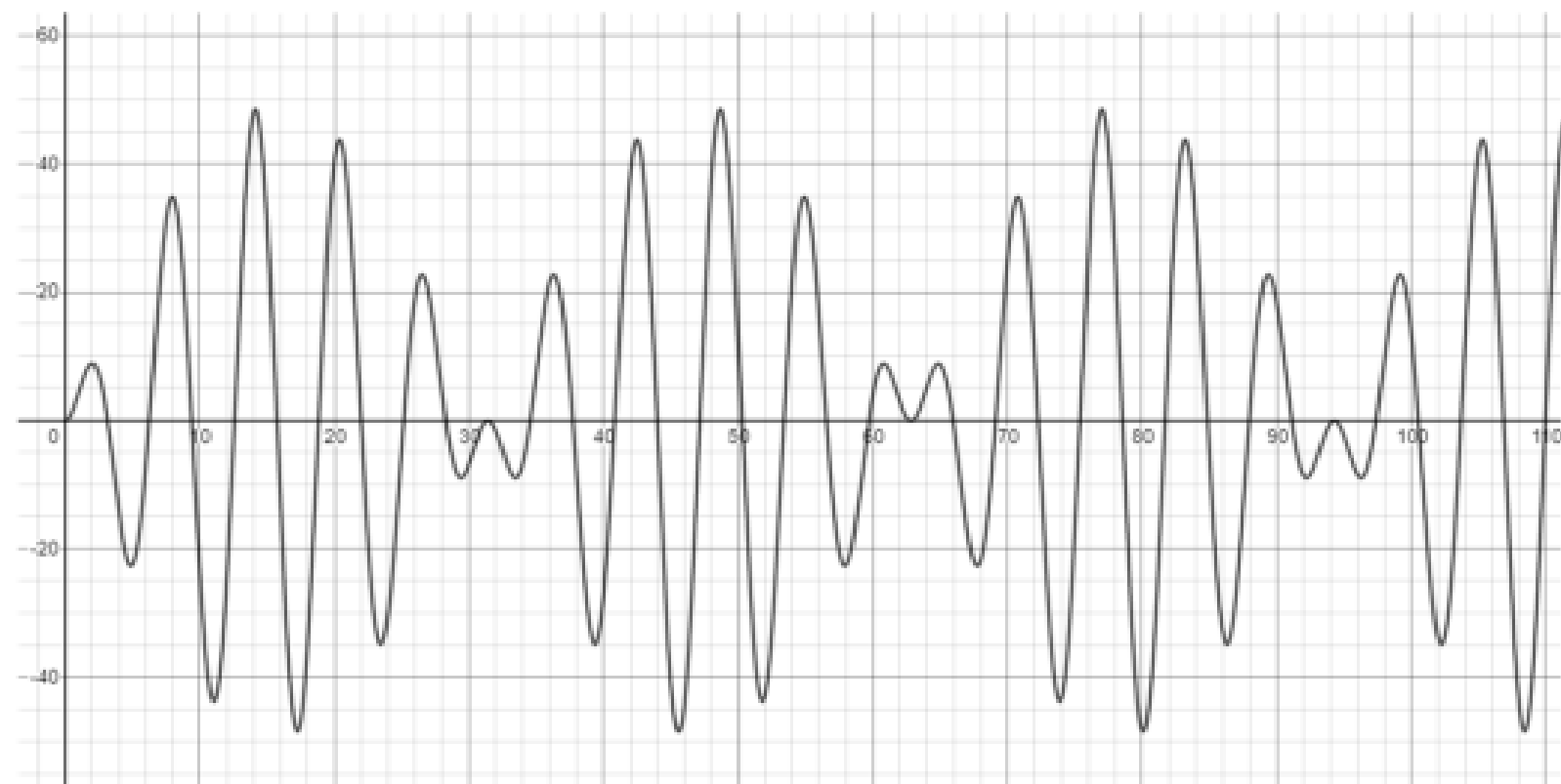
Suppose we have a particular solution (for a particular set of initial conditions) that is

$$y = \frac{F_0}{m(\omega_0^2 - \omega^2)} \cos(\omega t) - \frac{F_0}{m(\omega_0^2 - \omega^2)} \cos(\omega_0 t)$$

This can be rewritten as:

$$y = \frac{2F_0}{m(\omega_0^2 - \omega^2)} \sin\left(\frac{\omega_0 + \omega}{2}t\right) \sin\left(\frac{\omega_0 - \omega}{2}t\right)$$

When  $\omega_0$  and  $\omega$  are close, then the period of the last sin function is large. So we end up with something like:



# Beats Phenomenon

## Close Frequencies

When  $\omega$  is close to  $\omega_0$  but not equal

1

2

3

## Amplitude Modulation

Fast oscillation modulated by slow envelope function

## Solution Rewritten

$$y = \frac{2F_0}{m(\omega_0^2 - \omega^2)} \sin\left(\frac{\omega_0 + \omega}{2}t\right) \sin\left(\frac{\omega_0 - \omega}{2}t\right)$$

When the forcing frequency is close to but not equal to the natural frequency, we observe the beats phenomenon. This occurs when two waves of slightly different frequencies interfere, creating a pattern of alternating constructive and destructive interference.

Mathematically, we can rewrite our solution as a product of two sine functions. The first sine function represents a fast oscillation at the average of the two frequencies, while the second represents a slow modulation with a frequency equal to half the difference between the natural and forcing frequencies. This creates the characteristic "waxing and waning" pattern of beats.

## Case 2: Damped Forced Oscillations

Again the solution is a combination of the homogeneous solution and a particular solution.

But recall from the section on unforced oscillation that the unforced damped system will always tend to 0 as time increases. This means that:

'After a sufficiently long time the output of a damped vibrating system under a purely sinusoidal driving force will practically be a harmonic oscillation whose frequency is that of the input.'

So for damped forced oscillations, there is the **transient phase**, where the homogeneous solution will be involved, ie  $y = y_h + y_p$ , and the **steady-state phase**, where the homogeneous solution is negligible, so  $y = y_p$ .

In the damped case, we do not have 'run-away' resonance. Oscillations will always be finite (even theoretically), but some form of resonance still exists. This is called **practical resonance**.

The first thing we will note is that  $y_p(t) = a \cos(\omega t) + b \sin(\omega t)$  can be rewritten

$$y_p = C^* \cos(\omega t - \eta)$$

where  $C^*$  is called the **amplitude** and  $\eta$  is called the **phase angle**.

We can derive (see the textbook for details) that the **steady state amplitude** at a particular frequency  $\omega$  is given by:

$$C^*(\omega) = \frac{F_0}{\sqrt{m^2 (\omega_0^2 - \omega^2)^2 + \omega^2 c^2}}.$$

After some further derivation, we can find that the frequency at which the greatest amplitude occurs, called  $\omega_{max}$ , is given by

$$\omega_{max}^2 = \omega_0^2 - \frac{c^2}{2m^2}$$

with a corresponding amplitude of

$$C^*(\omega_{max}) = \frac{2mF_0}{c\sqrt{4m^2\omega_0^2 - c^2}}.$$

**Example:** Find the frequency  $\omega$  that would give the greatest amplitude of oscillation in a system modelled with the following differential equation

$$2y'' + 4y' + 7y = 4\cos(\omega t)$$

$$m=2, \quad c=4, \quad k=7, \quad F_0=4.$$

$$\text{natural frequency } \omega_0 = \sqrt{\frac{k}{m}} = \sqrt{\frac{7}{2}}$$

$$\text{driving frequency: } \omega_{\max}^2 = \omega_0^2 - \frac{c^2}{2m^2} = \frac{7}{2} - \frac{16}{8} = \frac{3}{2}$$

$$C^*(\omega_{\max}) = \frac{2mF_0}{c\sqrt{4m^2\omega_0^2 - c^2}}$$

$$= \frac{2 \times 2 \times 4}{4 \times \sqrt{4 \times 2^2 \times \frac{7}{2} - 4^2}} = \frac{4}{\sqrt{56 - 16}} = \frac{4}{2\sqrt{10}} = \frac{2}{\sqrt{10}}$$

In this lecture we have so far dealt with a mechanical forced mass - spring - damper system. There is a completely analogous electrical system: that of a resistor - inductor - capacitor (RLC) circuit with an applied voltage.

If we have a resistance  $R$ , an inductance  $L$ , a capacitance  $C$ , and an EMF of  $E_0 \sin(\omega t)$ , the current in the circuit,  $I$ , is modelled by

$$LI'' + RI' + \frac{1}{C}I = E_0\omega \cos(\omega t)$$

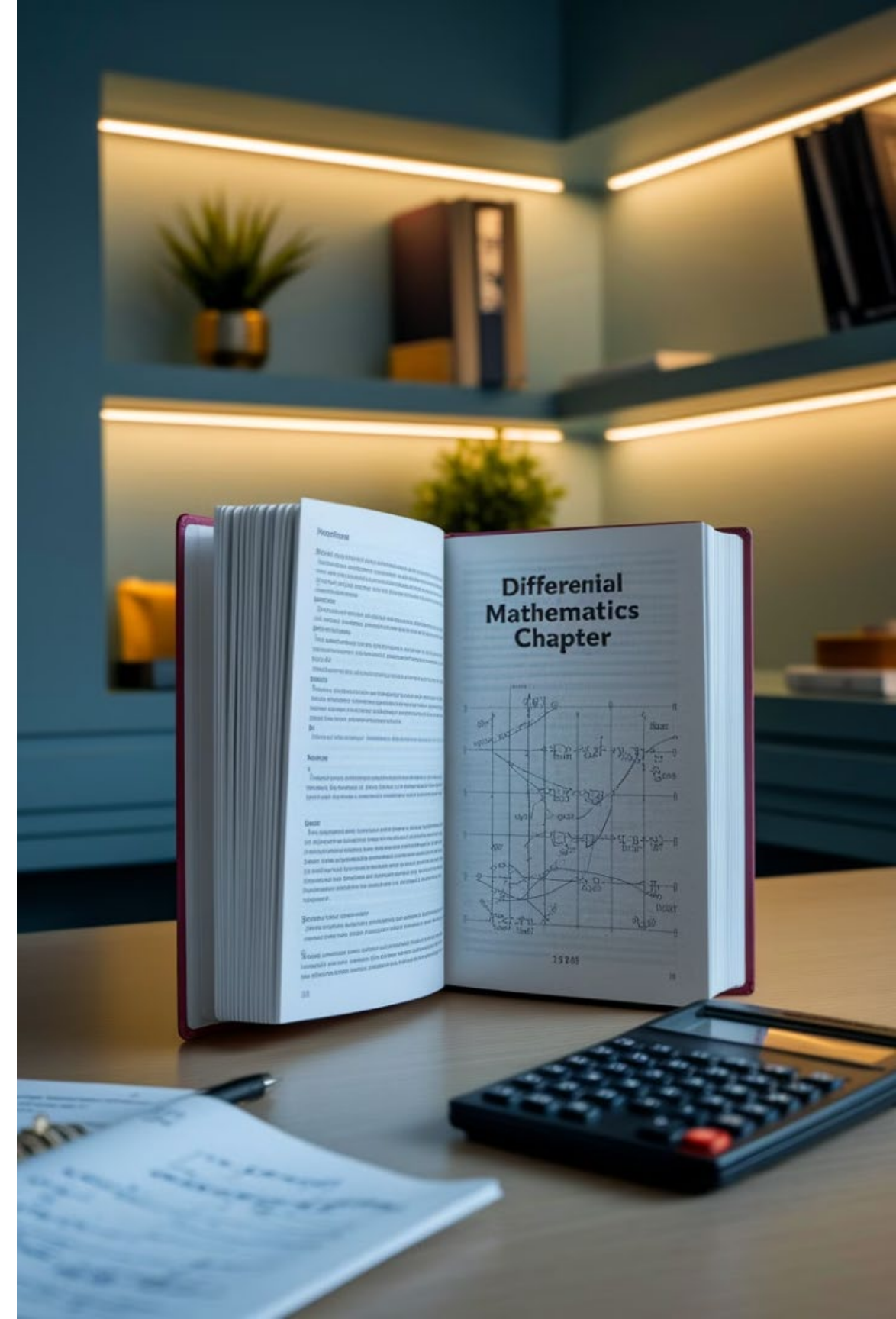
We will not go into the details of this in this lecture (it is covered in Chapter 2.9 in the textbook), but will note that there is a direct analogy between this model and the mechanical model we have studied today.

# Textbook References



## Chapter 2.8

Covers second order nonhomogeneous differential equations with constant coefficients





# What did you get out of today?

- **Purpose of today's lecture?** What was the purpose of of today's lecture?
- **Key learning:** What is one key thing you learnt?
- **Questions:** Is there anything that is still puzzling you?

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